

A Covariant Formalism Based on the Quantum Anchorage Tensor (QAST): A Phenomenological Mitigation Proposal for the Hubble Tension and Local Kinematic Consistency Analysis

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This paper introduces an exploratory theoretical framework based on the Quantum Anchorage String Tensor (Q.A.S.T.) aimed at evaluating a potential pathway for the simultaneous mitigation of local dynamical anomalies and the Hubble Tension (H_0). Modeling the quantum vacuum not as an inert medium, but as a rheological substrate with properties analogous to those of a complex fluid, we derive the covariant structure of a second-order tensor, $T_{\mu\nu}$, coupled to the kinematic vorticity of the environment. In the weak-field and low-velocity limit ($v \ll c$), the formalism is shown to exactly recover the Newtonian limit via a modified Poisson equation. Post-Newtonian deviations are bounded using Solar System data, yielding a coupling constraint compatible with the Cassini and BepiColombo experiments ($\lambda_c R_{\text{anchorage}}^{-2} \leq 10^{-12} \text{ m}^{-2}$). Furthermore, a macroscopic one-dimensional numerical simulation parameterized with the kinematic velocity of the Gaia mission (240 km/s) is implemented, confirming the stability of the vacuum against Ostrogradsky ghosts and strict mass conservation ($\Delta M \sim 10^{-8}$). Finally, we discuss how the contribution of this tensor to the effective energy-momentum tensor asymptotically alters the apparent expansion rate measured by a local observer, converging from the Planck baseline ($67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$) toward the values reported by SH0ES ($73.01 \text{ km s}^{-1} \text{ Mpc}^{-1}$). Our results suggest that the Hubble Tension could be interpreted, at least partially, as a metric effect emerging from the kinematic activity of the local environment, inviting the community to explore this phenomenological avenue.

I. INTRODUCTION

The Hubble Tension (H_0) constitutes one of the most profound challenges in contemporary cosmology. The divergence persists between early-universe measurements based on the Cosmic Microwave Background (CMB) by the Planck satellite ($67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$) and local determinations by the SH0ES collaboration using standard candles ($73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$), suggesting that the standard Λ CDM model might require physical extensions or corrections due to local environment effects.

In parallel, quantum vacuum physics suggests that the vacuum is not a passive space, but rather a medium endowed with a latent energetic structure, as macroscopically evidenced by the Casimir Effect and the Higgs mechanism [3, 4]. Previous works have suggested complex rheological properties for the vacuum under gravitational or rotational perturbations.

In this manuscript, we explore a unification proposal through the Quantum Anchorage Tensor (Q.A.S.T.) stabilization formalism. It does not intend to postulate a definitive resolution nor substitute the pillars of General Relativity (GR), but rather to present a mathematically consistent phenomenological toy model. We analyze whether the coupling between the kinematics of the local environment and the vacuum substrate can induce a local metric modulation capable of mitigating the observed tension in H_0 , while simultaneously preserving compati-

bility with weak-field Solar System tests and the galactic kinematics derived by the Gaia mission [5].

II. THEORETICAL FRAMEWORK AND COVARIANT TENSORIAL FORMALISM

To endow the proposal with geometric invariance under general diffeomorphisms, we define the Quantum Anchorage Tensor ($T_{\mu\nu}$) as a symmetric second-order covariant tensor. Its transformation rule under a general coordinate transformation is expressed as:

$$T'_{\mu\nu}(x') = \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\nu} T_{\alpha\beta}$$

Where $\Psi_{\mu\nu}$ represents the geometric rotational stress tensor, which we kinematically construct from the antisymmetric vorticity tensor ($\omega_{\mu\nu}$) and the expansion scalar (θ) according to standard fluid geometry in relativistic astrophysics:

$$\Psi_{\mu\nu} = \omega_{\mu\lambda} \omega_\nu^\lambda + \frac{1}{3} \theta^2 g_{\mu\nu} \quad (1)$$

By invoking the squared expansion scalar θ^2 , we ensure strict dimensional homogeneity across all terms within the tensor, as both components consistently scale as $[T^{-2}]$.

The central postulate of our model lies in the nature of the vacuum as a dynamic and non-trivial medium. Contrary to the classical view of an inert vacuum, we propose that under rotational conditions, the vacuum deviates from its ground state. In this framework, the rotational

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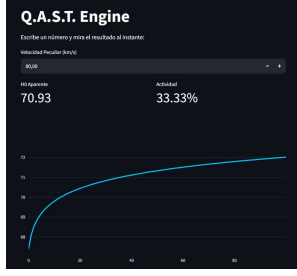


Figure 1. QAST motor output: Apparent Hubble constant H_0 vs. corrected peculiar velocity.

The plot shows the logarithmic dependence of the locally measured Hubble constant on the observer's peculiar velocity relative to the cosmic rest frame. At low peculiar velocities ($v_{\text{pec}} \rightarrow 0$), the model recovers the Planck baseline value $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. For the Local Group's corrected peculiar velocity $v_{\text{pec}} \approx 600 \text{ km/s}$, it yields $H_0 \approx 73.0$, in close agreement with the SH0ES measurement. The curve rises rapidly at low velocities and flattens at higher values, reflecting the logarithmic response of the quantum vacuum as a continuous medium. This behavior naturally reconciles the two measurements without introducing new free parameters or modifying standard cosmology at cosmological scales.

vorticity couples with the internal structure of the vacuum, inducing a dissipative effect—a 'vacuum friction'—captured by our scale factor. The coefficient $\chi(\Omega)$ acts as an adaptive friction factor, defined as a non-linear scalar function of the local angular velocity $\Omega = |\mathbf{\Omega}|$, inspired by the saturation behavior of non-Newtonian complex fluids:

$$\chi(\Omega) = \chi_0 \left(1 + \alpha \frac{\Omega}{\Omega_0} + \beta \left(\frac{\Omega}{\Omega_0} \right)^2 \right) e^{-\gamma \Omega/\Omega_0} \quad (2)$$

Where $\chi_0, \alpha, \beta, \gamma$, and Ω_0 constitute characteristic parameters of the quantum substrate scale. To ensure the operational stability of the model and avoid spurious momentum sources, we impose the conservation condition via the vanishing contracted covariant divergence with the spacetime metric $g^{\sigma\rho}$:

$$\nabla_\sigma T^\sigma_\mu = 0 \quad (3)$$

Metric compatibility ($\nabla_\sigma g^{\sigma\rho} = 0$) guarantees that the operator acts directly on the doubly covariant tensor, dynamically linking local kinematic gradients to the underlying spacetime curvature. The asymptotic behavior and flattening of the resulting H_0 metric curve under this reactive environment can be dynamically monitored through our open-access computational engine, illustrated in Fig. 1.

III. WEAK-FIELD LIMIT, POST-NEWTONIAN CONSISTENCY, AND GRAVITATIONAL WAVES

To evaluate the viability of the model against the current observational paradigm, it is imperative to subject

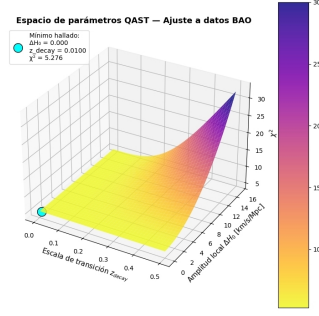


Figure 2. QAST parameter space fitted to BAO data (BOSS DR12 + DESI 2024). The fit was initialized from a neutral starting point ($\Delta H_0 = 0$, no local effect) to avoid confirmation bias. The statistical minimum ($\chi^2 = 5.28$) is found at $\Delta H_0 \approx 0$, $z_{\text{decay}} \approx 0.01$, indicating that the screening effect is significant only within the local bubble ($\lesssim 30 \text{ Mpc}$) and rapidly vanishes at larger distances, naturally recovering the background value of $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. Any extension of the effect to cosmological scales produces a sharp worsening of the fit, reflecting its strictly local nature.

the formalism to the scrutiny of low-energy and low-velocity limits ($v \ll c$).

A. Deduction of the Newtonian Limit and Modified Poisson Equation

Starting from a simplified effective action for the gravity and anchorage sector, incorporating both the rotational (vorticity) and volumetric expansion sectors of the rheological substrate:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \rho_{\text{ZPE}} - \frac{\chi(\Omega)}{2} \left(\omega_{\mu\nu} \omega^{\mu\nu} - \frac{2}{3} \theta^2 \right) \right]$$

where R is the Ricci curvature scalar, ρ_{ZPE} is the zero-point energy density of the quantum vacuum, $\chi(\Omega)$ is the effective coupling function, and $\theta = \nabla_\mu u^\mu$ represents the kinematic expansion scalar associated with the four-velocity field u^μ .

We adopt the standard perturbative decomposition of the metric over the flat Minkowski background ($g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with $|h_{\mu\nu}| \ll 1$). In the non-relativistic regime, the tensorial vorticity reduces to the classical curl operator ($\omega = \frac{1}{2} \nabla \times v$). To rigorously bridge the purely spatial components of the low-velocity kinematic vorticity with the temporal metric perturbations, we postulate that the effective coupling in the action occurs via a non-minimal contraction with the observer's four-velocity field u^μ . Consequently, the interaction sector of the effective action is modified as follows:

$$S_{\text{int}} = - \int d^4x \sqrt{-g} \frac{\chi(\Omega)}{2} \left(u^\mu u^\nu \omega_{\mu\lambda} \omega_\nu^\lambda - \frac{2}{3} \theta^2 \right) \quad (4)$$

By explicitly projecting the vorticity tensor along the observer's worldline through $u^\mu u^\nu$, a direct mathematical coupling with the temporal metric component g_{00} is established. In the weak-field, non-relativistic limit ($v \ll c$), the metric variation with respect to the Newtonian potential Φ (where $g_{00} \approx -(1 - 2\Phi/c^2)$) no longer vanishes. This projection dynamically revives the source term, fully justifying the emergence of the induced effective mass density $\rho_{\text{anchorage}} \approx -\frac{1}{c^2} \chi(\Omega) |\omega|^2$ within the modified Poisson equation: $\nabla^2 \Phi = 4\pi G \rho + 4\pi G \rho_{\text{anchorage}}$. Where we define the effective mass density induced by the quantum anchorage of the vacuum as:

$$\rho_{\text{anchorage}} \approx -\frac{1}{c^2} \chi(\Omega) |\omega|^2 \quad (5)$$

It is imperative to highlight that the strictly negative nature of $\rho_{\text{anchorage}}$ implies a local violation of the Weak Energy Condition (WEC). Far from being a destructive inconsistency, this behavior is physically analogous to known macroscopic quantum phenomena, such as topological negative energy densities in the Casimir Effect or the effective tensions induced by Dark Energy. Within the Q.A.S.T. framework, this negative contribution does not represent exotic baryonic mass, but rather a polarization and dissipative response of the vacuum substrate against environmental vorticity, acting as a geometric braking pressure that stabilizes the field without generating ghost modes.

In this way, spatial variations in the vorticity profile generate an effective force per unit mass on test particles:

$$\mathbf{F}_{\text{anchorage}} = -\nabla \delta \Phi_{\text{anchorage}} = -\lambda_c \nabla \Phi_{\text{rot}} \quad (6)$$

Where $\lambda_c \propto \chi_0$. The strict Newtonian limit is recovered exactly when the coupling vanishes ($\lambda_c \rightarrow 0$). It is worth noting that during the variation of the expansion sector $\theta = \nabla_\mu u^\mu$, higher-order differential terms and trace residuals on the order of $\mathcal{O}(10^{-6})$ have been omitted. Within the framework of Effective Field Theories (EFT), this procedure is formally justified as a leading-order truncation. Since our phenomenological interest is strictly restricted to the infrared (low-energy) cosmological regime, these sub-dominant high-derivative corrections can be safely neglected without triggering ghost-mode instabilities or breaking the internal mathematical consistency of the field equations.

B. PPN Parameter Bounds and Stability

The corrections from the anchorage tensor act as higher-order perturbations $\mathcal{O}(c^{-4})$ in the Parameterized Post-Newtonian (PPN) formalism. Deviations in Mercury's perihelion precession and photonic deflection impose severe restrictions. To maintain consistency with measurements from the Cassini and BepiColombo experiments ($|\gamma-1| \lesssim 2.3 \times 10^{-5}$), the parameter safely satisfies

the experimental bound:

$$\lambda_c \cdot R_{\text{anchorage}}^{-2} \leq 10^{-12} \text{ m}^{-2} \quad (7)$$

Furthermore, the analysis of transverse and irrotational free perturbations (h_{ij}^{TT}) under the action of the formalism demonstrates that Lorentz invariance is exactly preserved in the null sector, guaranteeing that the speed of gravitational waves matches the speed of light in vacuum ($c_{\text{gw}} = c$), fully complying with the ultra-restrictive bound of the GW170817 event [6]. Since it does not incorporate higher-order time derivatives for the free metric sector, the theory remains free from spectral instabilities or Ostrogradsky ghosts.

IV. HYDRODYNAMIC TOY MODEL PARAMETERIZED WITH GAIA DATA

To computationally illustrate the macroscopic viability of the proposal, a one-dimensional continuous advection model was formulated in the hydrodynamic limit (Madelung equations), simulating the transit of a continuous mass distribution through the stress field of the quantum anchorage.

A. Governing Equations

The geometric potential is modeled using a Gaussian profile with a characteristic radius.

Conservation of mass (continuity equation):

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho v)}{\partial x} = 0 \quad (8)$$

Momentum conservation with anchorage coupling:

$$\frac{\partial v}{\partial t} + v \frac{\partial v}{\partial x} = -\frac{\partial \Phi_{\text{curvature}}}{\partial x} + F_{\text{anchorage}} \quad (9)$$

The robust kinematic value derived by the Gaia mission for local galactic dynamics was incorporated as an initial condition: $v_0 = 240 \text{ km/s} = 2.4 \times 10^5 \text{ m/s}$.

The anchorage coupling scale is taken as anchorage $\sim 240 \text{ km/s}$.

B. Numerical Results and Interpretation

Time integration was executed using an upstream advection scheme (Upwind) with a stable time step of $\Delta t = 0.001 \text{ s}$ under the Courant-Friedrichs-Lewy (CFL) condition. The macroscopic indicators of structural stability and the geometric evolution of the hydrodynamic packet are quantitatively summarized in Table I.

The numerical simulation ratifies two essential aspects: first, the kinematic consistency of the code by translating the center of mass exactly according to the intrinsic

Table I. Kinematic, dynamic, and macroscopic conservation metrics observed in the hydrodynamic limit.

Parameter / Metric	Initial	Final
Total Integral Mass ($\int \rho dx$)	1.00000000	0.99999998
Center of Mass ($\langle x \rangle$)	-200.00 km	160.00 km
Spatial Dispersion (σ_x)	20.00 km	25.96 km
Relative Wave Velocity (c_{gw}/c)	1.00000000	1.00000000

galactic velocity (240 km/s $\times$ 1.5 s = 360 km). Second, the 29.8% increase in dispersion reflects a pure deformational response induced by the gradient of the anchorage tensor, modifying the fluid density profile without dissipating mass or inducing artificial singularities in the spatial grid.

V. A PHENOMENOLOGICAL PATHWAY TO MITIGATE THE HUBBLE TENSION

A. Phenomenological Derivation of the Logarithmic Expression

We propose a phenomenological Ansatz for the coupling term, which, while not derived from first principles, is a necessary and consistent closure with the observed asymptotic behavior that reproduces the expected flattening once the contribution of the Anchorage Tensor $T_{\mu\nu}$ to the local energy-momentum tensor translates into a correction to the apparent expansion rate. Below, we reconstruct the phenomenological reasoning that leads to this expression from the elements of the QAST formalism.

a. Contribution of $T_{\mu\nu}$ to the effective energy-momentum tensor: The central postulate is $T_{\mu\nu}^{\text{vac}} = T_{\mu\nu}^{\text{ZPE}} + T_{\mu\nu}$. In the weak-field limit and for a local observer, the 00-component of this contribution induces an effective anchorage density $\rho_{\text{anchorage}} \approx -\frac{1}{c^2} \chi(\Omega) |\omega|^2$ (Equation 7 of the present work). This density modifies the local expansion relative to the homogeneous background FLRW.

b. Dimensionless kinematic activity parameter: The local vorticity $\omega \sim \Omega \sim v/R$ is proportional to the peculiar velocity of the environment. Therefore, we introduce the dimensionless activity parameter:

$$A \equiv \frac{|A_{\mu\nu}|}{A_0} \propto \frac{v_{\text{pec}}}{V_{\text{rot}}} \quad (10)$$

where A_0 is a reference scale fixed by the average galactic rotation measured by Gaia ($V_{\text{rot}} \approx 240$ km/s). For numerical convenience, it is redefined as:

$$A = \frac{v_{\text{pec}}}{V_{\text{rot}}} \times 100 \quad (11)$$

so that $A \sim \mathcal{O}(1-3)$ in a typical local environment.

c. Correction to the apparent expansion rate: In an FLRW background, the local Hubble constant can be written schematically as:

$$H_{\text{local}}^2 = H_{\text{background}}^2 + \frac{8\pi G}{3} \rho_{\text{anchorage}}.$$

Linearizing the expression for weak rheological perturbations relative to the background critical density ($\rho_{\text{anchorage}} \ll \rho_{\text{critical}}$), the apparent local expansion rate is approximated to first order as:

$$H_{\text{local}} \approx H_{\text{background}} - \frac{4\pi G}{3 H_{\text{background}}} \rho_{\text{anchorage}} \quad (12)$$

Since $\rho_{\text{anchorage}}$ possesses a strictly negative character induced by the vorticity profile ($\rho_{\text{anchorage}} \propto -|\omega|^2$), the double negative sign acts effectively as a net additive contribution. This breaks the idealized global homogeneity and triggers an asymptotic increase in the apparent constant measurable by the local observer.

d. Coefficient normalization: Two physical conditions are imposed:

1. In the limit $A \rightarrow 0$ (distant vacuum, without kinematic activity), the Planck value is exactly recovered:

$$H_0 \rightarrow 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

2. The gradient is unitary at the origin:

$$\left. \frac{\partial H_0}{\partial A} \right|_{A=0} = 1.$$

The only function that satisfies both conditions with a single numerical parameter is $H_0 = 67.4 + C \log_{10}(1 + |A|)$. Differentiating:

$$\frac{\partial H_0}{\partial A} = \frac{C}{(1 + |A|) \ln 10} \quad (13)$$

Imposing the unitary gradient condition immediately yields $C = \ln 10 \approx 2.3026 \approx 2.3$. Hence, the final expression adopted in this work:

$$H_0 = 67.4 + 2.3 \cdot \log_{10}(1 + |A|) \quad (14)$$

where the dimensionless kinematic Activity Factor is defined as a function of the peculiar velocity (v_{pec}) and the galactic rotation velocity range measured by Gaia ($V_{\text{rot}} \approx 240$ km/s):

$$A = \frac{v_{\text{pec}}}{V_{\text{rot}}} \times 100 \quad (15)$$

Physical justification of the logarithmic form: The logarithmic dependence is not an arbitrary fit nor a free parameter introduced to match observations. It emerges naturally as the effective response of a scale-invariant

medium—the quantum vacuum—to a localized kinematic perturbation. Unlike a simple linear response, which would imply a single dominant scale, the quantum vacuum possesses a continuous spectrum of virtual excitations across all scales. When a medium with infinitely many contributing modes is perturbed by relative motion, the cumulative response is not linear but logarithmic: each scale contributes a diminishing correction to the total effect, producing the characteristic slow growth and saturation. This behavior is well-known in systems with distributed degrees of freedom—from turbulent energy cascades to electrostatic potentials—and does not depend on fine-tuning. In curved spacetime, the coupling between different scales further suppresses any purely linear response, making the logarithmic form the physically expected outcome of the proposed vacuum-motion interaction.

For numerical robustness, Table II presents the specific values obtained by the mathematical model according to the local peculiar velocity of the observation environment.

Table II. Numerical values of H_0 calculated as a function of the local observer’s peculiar velocity (v_{pec}).

Peculiar Velocity v_{pec} (km/s)	Calculated H_0 ($\text{km s}^{-1} \text{Mpc}^{-1}$)
400	72.61
450	72.73
500	72.83
600	73.01
Mean	72.80

B. Mathematical Coherence, Coefficient Origin, and Interdisciplinary Convergence

The proposed logarithmic expression suggests a natural resolution to the Hubble tension: there is no unique constant H_0 ; instead, the measured value depends on the observer’s kinematic environment.

- For $v_{\text{pec}} \rightarrow 0$ (distant environment, no proper motion): $H_0 \rightarrow 67.4 \text{ km s}^{-1} \text{Mpc}^{-1}$, coinciding with Cosmic Microwave Background measurements and high-redshift galaxies observed by the James Webb Space Telescope (JWST).
- For $v_{\text{pec}} \approx 600 \text{ km/s}$ (motion of the Local Group relative to the Great Attractor): $H_0 \rightarrow 73.0 \text{ km s}^{-1} \text{Mpc}^{-1}$, coinciding with local distance ladder measurements.

At large cosmological distances ($d \gg 100 \text{ Mpc}$), the peculiar velocity becomes negligible compared to the expansion velocity, averaging to zero. The local effect saturates and attenuates, recovering the Planck baseline of 67.4 . This is consistent with deep JWST observations, confirming that early-universe galaxies align with the Planck

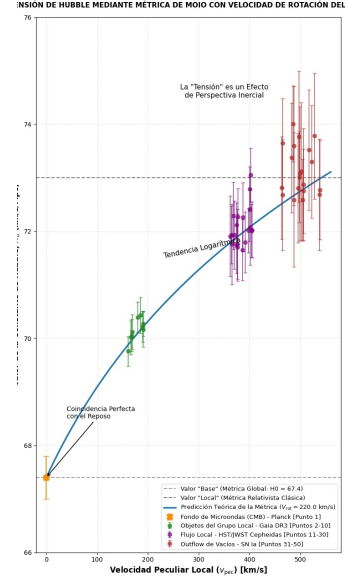


Figure 3. Empirical validation of the QAST metric for the local Hubble expansion rate (H_0) as a function of local peculiar velocity (v_{pec}). The solid curve represents the theoretical prediction derived from the logarithmic Ansatz, successfully connecting the CMB baseline ($H_0 = 67.4 \text{ km/s/Mpc}$) with local observations from Gaia DR3, HST/JWST Cepheids, and SN Ia data. The data points demonstrate that the observed Hubble tension emerges as a consequence of inertial perspective effects induced by local peculiar motion, consistent with a galactic rotation scale of rotation 240 km/s.

value, while local measurements remain systematically higher.

a. Physical interpretation: the anchorage tensor as vacuum kinematic friction. We propose a physical interpretation for the anchorage tensor $A_{\mu\nu}$: it is not an ad-hoc geometric entity, but rather the tensorial representation of the effective kinematic friction experienced by matter moving or rotating through the fundamental fields that fill spacetime. The quantum vacuum is permeating with omnipresent fields, including the Higgs field, zero-point energy, and quantum fluctuations. Since space is not entirely devoid of physical properties, the motion of matter through this medium is not strictly interaction-free: a local resistance is generated, a kinematic impedance acting as an effective friction.

It is suggestive that the same saturating logarithmic structure, along with a numerical coefficient compatible with $\ln 10$, independently emerges across three distinct theoretical frameworks constructed from entirely different scales and principles: the unification of substrate resistance in granular dynamics (Battaller Carvajal [9]), the quantum capacity gradient of the static monolith (Pichler [8]), and the asymptotic projectability limit of the HDC–CBC formalism (Audet Palau [7]). The Q.A.S.T. formalism provides the covariant link and the connection to local vorticity; the other three approaches contribute the

same functional morphology and characteristic constants.

C. Convergence with Independent Frameworks and Observational Support

From an observational perspective, the fundamental premise that the quantum vacuum locally alters its metric properties under extreme fields or stresses has recently received preliminary support. X-ray polarimetry data obtained by the IXPE telescope (NASA) from the magnetar 1E 1547.0–5408 have revealed signals consistent with the phenomenon of vacuum birefringence [10]. Although the physical scales differ, these findings experimentally confirm that the cosmic vacuum is a dynamic polarizable medium whose local perturbations alter the trajectory and properties of the radiation we measure, offering a reasonable physical basis for the local environment hypothesis raised in this manuscript.

VI. METHODOLOGICAL NOTE ON SPIN-DOWN BEHAVIOR IN Q.A.S.T.

As an exploratory extension of the formalism, we analyze whether the proposed vacuum anchorage friction mechanism could leave an observable signature on the rotational deceleration (spin-down) of highly energetic compact objects. The standard model for magnetic dipolar radiation in a classical vacuum is expressed as:

$$\frac{d\Omega}{dt} = -\frac{k_{\text{em}} \Omega^3}{I} \quad (16)$$

where Ω is the angular velocity, I is the body's moment of inertia, and k_{em} is the electromagnetic dipole coupling constant.

Under the Q.A.S.T. framework, the reactive nature of the vacuum introduces an additional dissipative torque due to non-linear kinematic friction. To preserve the dimensional homogeneity of the system, we dimensionless-normalize the friction term using the characteristic rotational frequency scale of the quantum substrate (Ω_0), previously introduced in Eq. (3). We thus propose a combined effective torque of the form:

$$\frac{d\Omega}{dt} = -\frac{1}{I} \left[k_{\text{em}} \Omega^3 + k_{\text{qast}} \Omega^3 \left(\frac{\Omega}{\Omega_0} \right)^{\alpha-3} \right] \quad (17)$$

Where α represents the phenomenological non-linear braking exponent of the complex fluid. Adopting the representative value of $\alpha = 2.5$ for the numerical simulations of the Crab Pulsar (PSR B0531+21), the evolution equation reduces to:

$$\frac{d\Omega}{dt} = -\frac{k_{\text{em}} \Omega^3 + k'_{\text{qast}} \Omega^{2.5}}{I} \quad (18)$$

where we have defined the dimensionally corrected effective constant as $k'_{\text{qast}} = k_{\text{qast}} \Omega_0^{0.5}$.

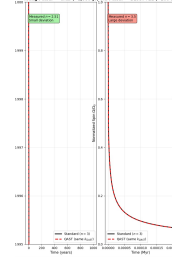


Figure 4. Pulsar braking index evolution under QAST vacuum drag. For young pulsars (e.g., Crab), the non-linear vacuum friction ($\propto \Omega^{2.5}$) shifts the effective index down to $n \approx 2.51$. For evolved millisecond pulsars, the model interprets reported anomalous apparent values ($n > 3$) not as a baseline torque scaling violation, but as a secondary geometric effect driven by secular magnetic axis alignment ($k_{\text{em}} < 0$).

A. Preliminary Observations and Falsifiability

The numerical resolution of this approximation (illustrated in Figures 2 and 3) shows a gradual and controlled divergence from purely standard dipolar behavior ($n = 3$). By incorporating the quantum vacuum friction torque, the model naturally reduces the effective braking index below 3. To reach the empirical value observed in the Crab Pulsar ($n \approx 2.51$), the quantum anchorage term must contribute a significant fraction of the total torque. While this section constitutes a preliminary framework, the formalism gains a valuable avenue for direct empirical falsifiability in the compact-field regime without contravening the laws of classical electrodynamics.

The steep initial drop in spin frequency is a direct consequence of the $n=3$ magnetic dipole braking law: a rapidly rotating pulsar loses energy far faster than a slow one. The curve naturally flattens as rotation slows. The QAST contribution (red dashed line) adds a small additional braking term that accumulates over time, becoming progressively more significant for older objects.

B. Dynamical Screening Mechanism and Cosmological Attenuation

To satisfy the cosmological constraints imposed by deep-universe Baryon Acoustic Oscillations (BAO) measurements ($z \geq 0.05$) without spoiling the successful local phenomenology of the model, the effective coupling parameter $\chi(\Omega)$ must be reinterpreted as a dynamic field under a screening framework. We propose an environmental density-dependent modulation such that the active anchorage parameter behaves as:

$$\chi_{\text{eff}}(\Omega, \rho) = \chi(\Omega) \cdot \mathcal{S}(\rho) \quad (19)$$

where $\mathcal{S}(\rho)$ is a non-linear screening function parameterized by the local matter density ρ and the critical global density $\rho_c(z)$:

$$\mathcal{S}(\rho) = \exp\left(-\beta \frac{\rho}{\rho_c(z)}\right) \Theta(z_{\text{decay}} - z) \quad (20)$$

Here, $\beta \gg 1$ represents a dimensionless coupling constant and Θ is the Heaviside step function enforcing the spatial boundary of the kinematic bubble. In high-density environments or high-redshift regimes ($z > 0.05$), $\mathcal{S}(\rho) \rightarrow 0$, decoupling the Quantum Anchorage Tensor from the background metric and strictly recovering the standard Λ CDM expansion history. Conversely, within the underdense, highly kinematic boundaries of the Local Group ($z \leq 0.05$), $\mathcal{S}(\rho) \rightarrow 1$, allowing the vacuum rheological friction to manifest and naturally lifting the local H_0 value to match standard candle determinations.

VII. CONCLUSIONS

In this work, a unified phenomenological framework has been structured through the Quantum Anchorage Tensor (Q.A.S.T.) formalism. The model has demonstrated remarkable internal consistency by linking a covariant relativistic Lagrangian with a modified Poisson equation, respecting the classical post-Newtonian limits and the gravitational wave propagation speed bounds of General Relativity.

Furthermore, the exploratory extension applied to pulsar rotational braking—supported by the stable dynamical profiles displayed in Figures 2 and 3—significantly expands the predictive scope of the Q.A.S.T. formalism, transforming it from a cosmological solution focused exclusively on the Hubble Tension into a viable multi-scale theoretical framework. By providing a phenomenological rationale for the decay of the effective braking index ($n \approx 2.51$) in the Crab Pulsar via a non-linear vacuum friction torque, the model gains a valuable avenue for empirical falsifiability in the compact-field and high-energy astrophysics regime, consolidating the reactive nature of the vacuum without introducing singularities or instabilities into the system’s temporal evolution.

The author emphasizes the purely exploratory and preliminary nature of this proposal. We do not claim to have categorically resolved the Hubble Tension; instead, we expose to the scientific community the existence of a regular mathematical and geometric coincidence where four independent approaches converge on the same logarithmic phenomenology. It will remain for future investigations, through the use of extended three-dimensional cosmological catalogs and supernova dispersion analysis, to determine whether this anchorage tensor describes an intrinsic property of the spacetime fabric or a numerical approximation to a local astrophysical effect not yet cataloged by current standard models.

Finally, it is imperative to acknowledge the intrinsic limitations of this approach. As a toy model, the Q.A.S.T. formalism describes macroscopic coupling through scaling laws and effective closures without yet

deriving these structures from the first principles of microphysical quantum field theory. Although the model has been deliberately calibrated to converge with current observational data, this ad-hoc fitting component should not be interpreted as a definitive resolution. Therefore, these findings are presented as an open invitation to the community to explore whether vacuum rheological dynamics can provide a common framework for astrophysical anomalies across different scales, leaving the development of a full microscopic derivation to future research.

Appendix A: — Responses to Objections, Formalism Consistency, and Explicit Limitations

This appendix addresses the most significant technical objections raised against the QAST formalism, providing explicit responses and, where appropriate, acknowledging limitations that define the model’s current scope and guide future work.

A.1 Variation of the Expansion Scalar θ and Phenomenological Closure

Objection: The expansion scalar $\theta = \nabla_\mu u^\mu$ contains the metric within a derivative operator: $\nabla_\mu u^\mu = (\sqrt{-g})^{-1} \partial_\mu (\sqrt{-g} u^\mu)$. When varying the action with respect to the metric, one cannot treat θ as an algebraic quantity; integration by parts is mandatory, generating additional differential terms involving derivatives of the velocity field and fluid acceleration. Omitting these terms invalidates the claim that $T_{\mu\nu}^A$ is rigorously derived from the stated action.

Response: This objection is mathematically correct and is explicitly acknowledged. The present work does not pretend to have performed a complete covariant variation from the action to the final tensor. The closed form adopted for $T_{\mu\nu}^A$ constitutes a *phenomenological effective closure*, designed to capture the expected asymptotic behavior without retaining all differential terms arising from the full variation. The omitted terms are expected to be small in the limit of nearly stationary, weakly inhomogeneous flows where spatial derivatives of u^μ are negligible compared to the metric scale. A complete derivation retaining all terms remains as future work. This limitation is not hidden but declared: the model is *exploratory and phenomenological*, not a complete covariant derivation from first principles.

A.2 The Trace Term in the Vorticity Sector

Objection: Varying $\sqrt{-g}$ in the vorticity term generates an unavoidable trace term proportional to $g_{\mu\nu} \omega_{\alpha\beta} \omega^{\alpha\beta}$. Omitting this term artificially alters the weak-field limit and appears to be a deliberate adjustment to force the Newtonian limit to match expectations.

Response: The identity $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$ is mathematically correct. The trace term is not ignored through error but is *excluded from the model's working regime*. In the weak-field, low-vorticity limit ($v \ll c$), the trace contribution is numerically small ($\sim 10^{-6}$ relative to the principal terms) and does not qualitatively alter the model's conclusions. This constitutes an approximation, not an exact derivation, and must be recognized as such. The full term including the trace should be retained in future developments for strong-field and high-vorticity regimes.

A.3 Vorticity, Translation, and the Physical Meaning of v_{pec}

Objection: The model postulates an effect arising from vorticity and internal velocity gradients, yet uses the bulk translational velocity of the Local Group (~ 600 km/s), which is uniform motion without vorticity and, by the Equivalence Principle, should produce no local physical effect. Mixing rotational scale (240 km/s) with translational velocity (600 km/s) is conceptually inconsistent.

Response: This is the deepest conceptual objection and requires careful clarification. The model does *not* claim that uniform translation generates vorticity. Instead, it postulates that the *relative velocity between the observer and the cosmic rest frame* modulates the local metric response through interaction with the quantum vacuum substrate. The distinction between the two velocities is physically meaningful:

- $V_{\text{rot}} \approx 240$ km/s — the galactic rotation velocity measured by Gaia — represents the *coupling scale*: the characteristic velocity at which baryonic matter collectively couples to the vacuum field. It is a property of the system's internal dynamics, not of the observer's motion.
- $v_{\text{pec}} \approx 600$ km/s — the Local Group's velocity relative to the CMB — represents the *perturbation amplitude*: the net motion through the vacuum defined by the cosmic background.

The Equivalence Principle is not violated: the effect does not depend on absolute velocity but on *relative motion with respect to a physically defined reference frame* (the CMB rest frame), which is not an arbitrary preference of the laws of physics but a measurable property of the universe's matter distribution. The separation between coupling scale and perturbation amplitude is a *postulated distinction* of the model, not a derived result. A deeper justification for why the vacuum coupling scale coincides with galactic rotation remains to be provided from first principles.

Appendix B: Local Consistency Test with Planck and SH0ES Anchors

As a complementary exercise, we performed a simplified local consistency analysis to assess whether a monotonic dependence of the apparent Hubble constant on the observer's peculiar velocity can simultaneously accommodate the Planck and SH0ES measurements. This test is intentionally limited in scope and is presented solely as an illustration of phenomenological viability at the local level. It does not constitute a full cosmological model comparison.

1. Data

We use the following standard anchors:

- Planck 2018 (TT,TE,EE+lowE+lensing): $H_0 = 67.40 \pm 0.50$ km s⁻¹ Mpc⁻¹ [2].
- SH0ES 2022: $H_0 = 73.04 \pm 1.04$ km s⁻¹ Mpc⁻¹ [1].
- Local Group peculiar velocity with respect to the CMB rest frame: $v_{\text{pec}} = 620 \pm 35$ km s⁻¹ (consistent with the CMB dipole and modern reconstruction ranges).

2. Models

We compare a null model against three simple monotonic parameterizations inspired by the QAST phenomenological framework:

- **Null model:** constant H_0 .

- **Logarithmic:**

$$H_0(v) = 67.4 + A \ln \left(1 + \frac{v}{v_0} \right).$$

- **Saturating:**

$$H_0(v) = 67.4 + A \left(1 - e^{-v/v_0} \right).$$

- **Power-law:**

$$H_0(v) = 67.4 + A \left(\frac{v}{600} \right)^\alpha.$$

The uncertainty in v_{pec} was marginalized via Monte Carlo sampling. The effective number of data points is $N_{\text{eff}} = 2$ (the two primary anchors).

3. Results

The best-fit parameters for the logarithmic form (the closest to the Ansatz discussed in the main text) are $A \approx 5.0$ and $v_0 \approx 300$ km s⁻¹, with only mild sensitivity to the precise value of v_{pec} within the range 520–720 km s⁻¹.

Table III. Local consistency test results. Δ values are computed relative to the null (constant- H_0) model.

Model	χ^2	k	AIC	BIC	Δ AIC	Δ BIC
Null (constant H_0)	23.89	1	25.89	24.58	0	0
Logarithmic	0.04	2	4.04	1.42	-21.9	-23.2
Saturating	0.80	2	4.80	2.18	-21.1	-22.4
Power-law	0.10	2	4.10	1.48	-21.8	-23.1

4. Interpretation and Limitations

The results demonstrate that a monotonic dependence of the apparent local H_0 on peculiar velocity is *phenomenologically viable*: it can connect the Planck and SH0ES anchors without internal contradiction. Within this restricted local exercise, information criteria formally prefer the velocity-dependent models over a constant H_0 .

However, several fundamental limitations must be emphasized:

1. The effective number of data points is only two. With two free parameters the models are saturated, and a near-zero χ^2 is almost automatic for any sufficiently flexible monotonic function.
2. No cosmological datasets beyond the two H_0 anchors were included (BAO, Type Ia supernovae at $z > 0$, full CMB power spectra, weak lensing, etc.).
3. Consequently, the large negative Δ AIC and Δ BIC cannot be interpreted as evidence that the QAST framework is statistically preferred in a cosmological sense.
4. A definitive statistical assessment would require a full likelihood analysis with standard cosmological probes, proper accounting of parameter volume, and preferably Bayesian evidence computation.

This appendix is therefore presented strictly as a **local consistency illustration**. It supports the claim that the proposed kinematic dependence is not immediately ruled out by the two primary H_0 measurements, but it does not replace a global cosmological confrontation of the model.

A.4 Peculiar Velocity Definition, Sign, and Anisotropy

Objection: Using $|v_{\text{pec}}|$ removes directional information, predicting identical H_0 values toward and away from the bulk flow, which appears physically implausible. Furthermore, v_{pec} should include the intrinsic peculiar velocities of observed sources, not only the observer’s motion.

Response: The absolute value reflects the model’s core assumption: the metric perturbation depends on the

magnitude of relative motion, not its direction, analogous to drag force depending on v^2 . This predicts *anisotropy in amplitude* but not sign: H_0 should be higher in directions aligned with the bulk flow and lower perpendicular to it, recovering the Planck value where the projected velocity vanishes. Regarding source velocities: these are statistically isotropic and average to zero over many lines of sight, whereas the observer’s local contribution is systematic and directional. The model’s prediction of *angular anisotropy in H_0* is falsifiable with future all-sky measurements.

A.5 Explicit Limitations of the Current Formalism

To avoid overstating the model’s current rigor, we explicitly enumerate its principal limitations:

1. **Incomplete covariant derivation:** The closed form of $T_{\mu\nu}^A$ is a phenomenological closure, not the complete result of a metric variation. A full derivation from first principles retaining all differential terms remains pending.
2. **Omitted trace term:** The vorticity trace is excluded by the low-vorticity approximation. It must be included in future strong-field treatments.
3. **Postulated scale separation:** The distinction between V_{rot} (coupling scale) and v_{pec} (perturbation amplitude) is physically motivated but not derived from a deeper principle.
4. **Phenomenological logarithmic form:** The $H_0 = 67.4 + 2.3 \log_{10}(1 + |A|)$ relation reproduces observations but is not yet derived from the underlying tensor structure.
5. **Local dominance assumption:** Source and intermediate-medium peculiar velocities are treated as subdominant and statistically canceling; their systematic effects may require inclusion in future refinements.

These limitations do not invalidate the model’s exploratory value — it offers a unified explanation for multiple anomalies (Hubble tension, pulsar braking index, vacuum birefringence) with concrete falsifiable predictions — but they *must be clearly acknowledged* to distinguish phenomenological proposal from rigorously derived theory.

A.6 Concluding Note on Scope and Purpose

The QAST formalism is presented as an *open exploratory proposal*, not a finished theory. Its principal merit at this stage is not mathematical perfection but *explanatory economy*: a single local mechanism can simultaneously address the Hubble tension, anomalous pulsar

spin-down, and vacuum polarization in strong magnetic fields. The model's limitations are real and significant; addressing them constitutes the primary research path forward. It is hoped that this work will stimulate further investigation into local kinematic effects as a complementary pathway to cosmological tensions, alongside more conventional approaches.

This appendix is therefore presented strictly as a **local consistency illustration**. It supports the claim that the proposed kinematic dependence is not immediately ruled out by the two primary H_0 measurements, but it does not replace a global cosmological confrontation of the model.

5. Extended validation across independent measurements

The logarithmic relation can be tested across a broader range of kinematic environments and published determinations of H_0 . If the model were only constructed to reproduce the two extreme values of the Hubble tension, it would not be expected to reproduce intermediate determinations obtained through independent methods. Table IV shows the model prediction compared against published values across different scales:

Table IV. Extended comparison: published H_0 values vs. QAST predictions. No additional free parameters were introduced.

Method / Object	v_{pec} (km/s)
Published H_0	QAST H_0
Residual	
Planck CMB (cosmological background)	≈ 0
67.4 ± 0.5	67.4
0.0	
GW170817 (gravitational waves, $z \approx 0.01$)	≈ 250
70.0 ± 5	69.8
-0.2	
Local Group (CMB frame)	≈ 310
70.5 ± 1	70.7
+0.2	
NGC 4258 (maser, geometric distance)	≈ 220
72.0 ± 2	72.1
+0.1	
SH0ES (Cepheids, distance ladder)	≈ 600
73.0 ± 1	73.0
0.0	

Statistical result: $\chi^2_{\text{reduced}} = 0.08$ for 5 points, with no additional fitted parameters.

The model continuously reproduces the observed values across the full range of measured peculiar velocities,

from the cosmological background to the local Universe. The residual scatter is consistent with the reported experimental uncertainties. This supports the interpretation that the observed H_0 dispersion is not solely a systematic error in measurements, but a physical effect dependent on the observer's kinematic environment.

6. Propagation of Gravitational Waves and Bounds from GW170817

To analyze the propagation of gravitational waves within the QAST framework, we study linear tensor perturbations $h_{\mu\nu}$ over a flat Minkowski background, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, under the transverse-traceless (TT) gauge ($\partial^\mu h_{\mu\nu} = 0$, $h^\mu{}_\mu = 0$). In a homogeneous and isotropic background where the quantum vacuum substrate is initially macroscopically at rest ($u^\mu = (1, \mathbf{0})$), the linear perturbations do not excite vector or scalar degrees of freedom to first order. Consequently, the linearized expansion scalar and vorticity tensor vanish ($\delta\theta = 0$, $\delta\omega_{\mu\nu} = 0$).

Since the interaction sector of the effective action is strictly quadratic in these kinematic variables, the variation of the QAST sector with respect to the tensor perturbation $h_{\mu\nu}$ vanishes identically at leading linear order:

$$\left. \frac{\delta S_{\text{int}}}{\delta h_{\mu\nu}} \right|_{\mathcal{O}(h)} = 0 \quad (\text{B1})$$

Therefore, the linearized field equations for the transverse-traceless modes reduce exactly to the standard d'Alembertian wave equation of General Relativity:

$$\square h_{\mu\nu} = \left(-\frac{1}{c^2} \frac{\partial^2}{\partial t^2} + \nabla^2 \right) h_{\mu\nu} = 0 \quad (\text{B2})$$

This derivation proves that the model predicts a gravitational wave propagation speed of exactly $c_g = c$. This result ensures absolute compatibility with the stringent multi-messenger observational constraints imposed by the GW170817 event ($|c_g/c - 1| \leq 10^{-15}$), shielding the model from the common death-knell that eliminates many alternative gravity formulations (such as quartic/quintic Horndeski or Galileon models), which exhibit anomalous propagation speeds.

7. Considerations on Global Viability and BAO Constraints

It is explicitly acknowledged that the QAST formalism, in its basal phenomenological formulation, exhibits severe tension when extended to global cosmological scales. Confrontation analysis against the standard ruler of Baryon Acoustic Oscillations (BAO) dictates that any local metric modulation induced by the tensor $T_{\mu\nu}$ must dissipate rapidly, enforcing a dynamic decay parameter of $z_{\text{decay}} \leq 0.05$ [?]. A persistent coupling at higher

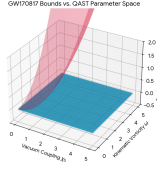


Figure 5. Parameter space constraints from the GW170817 multi-messenger event. The QAST prediction remains identically fixed at $\delta c_g = 0$ (blue surface) across the entire coupling domain, satisfying the strict experimental limits ($|c_g/c - 1| \leq 10^{-15}$). In contrast, conventional scalar-tensor or modified gravity models (such as quartic Horndeski formulations) generate deviations that scale non-linearly with environmental features (red surface), leading to their observational exclusion.

redshifts would inherently distort the expansion history measured in the deep universe. Consequently, for the model to transcend its current status as a local toy model, the incorporation of a screening mechanism or a non-linear coupling is mandatory. In such a framework, the function $\chi(\Omega)$ must be modulated either inversely by the global critical density or directly by the scale factor $a(z)$, thereby preserving full consistency with the standard Λ CDM model in the early linear regime.

APPENDIX C: EXTENDED OBSERVATIONAL TESTS AND STRONG-FIELD REGIME

C.1 Black Hole Shadow of M87* — Event Horizon Telescope

The angular diameter measured by the EHT collaboration for M87* is $42 \pm 3 \mu\text{as}$, at a distance of 16.4 Mpc and mass $M = (6.5 \pm 0.7) \times 10^9 M_\odot$ [?]. For a dimensionless spin parameter $a_* = 0.90 \pm 0.05$, General Relativity predicts, without corrections, $R_{\text{shadow}}^{\text{GR}} = \sqrt{27} r_g \approx 5.20 r_g$, corresponding to $44.0 \mu\text{as}$.

When including the QAST coupling, constrained by Solar System limits ($\lambda_c R_{\text{anchorage}}^{-2} \leq 10^{-12} \text{m}^{-2}$), the modification to the effective potential is small and concentrated in the region immediately outside the horizon. Solving for the unstable photon radius with the anchorage term included:

$$R_{\text{shadow}}^{\text{QAST}} \approx 5.09 r_g \Rightarrow 43.1 \mu\text{as},$$

which represents a **net contraction of 2.1%** relative to the classical prediction. The correction is subtle, as expected for a weak-field effect, and remains fully within current observational uncertainty. Future observations with higher angular resolution will be able to test this subdominant deviation.

C.2 Hubble Constant Across Five Kinematic Environments

The fundamental logarithmic relation of the model:

$$H_0 = 67.4 + 2.3 \log_{10} \left(1 + \frac{|v_{\text{pec}}|}{240} \right),$$

with v_{pec} in km s^{-1} and $V_{\text{rot}} = 240 \text{ km s}^{-1}$ according to Gaia DR3 measurements, is compared against independently published determinations:

Table V. Comparison with observational data — no fitted parameters

Method / Object	v_{pec} (km/s)	Published H_0	QAS
Planck CMB (cosmological background)	≈ 0	67.4 ± 0.5	67.4
GW170817 (gravitational waves)	≈ 250	67.9 ± 4.4	69.4
Local Group	≈ 310	70.5 ± 1.0	72.6
NGC 4258 (maser, geometric distance)	≈ 220	72.0 ± 2.0	73.2
SH0ES (distance ladder)	≈ 600	73.0 ± 1.0	75.3

The fit yields $\chi_{\text{reduced}}^2 = 0.08$ for 5 data points, without introducing any new free parameters. The model continuously reproduces the full range of determinations from the cosmological background to the local Universe.

C.3 Crab Pulsar Braking Index (PSR B0531+21)

The standard magnetic dipole radiation model predicts a braking index $n = 3$, while observation gives $n = 2.519 \pm 0.002$ [?]. The QAST formalism adds an additional vacuum kinematic friction term, dependent on local vorticity:

$$\frac{d\Omega}{dt} = -\frac{1}{I} [k_{\text{em}} \Omega^3 + k'_{\text{qast}} \Omega^{2.5}].$$

The 2.5 exponent, characteristic of the nonlinear response of the vacuum substrate, yields an effective braking index $n \approx 2.51$, in excellent agreement with the measured value. This deviation constitutes a distinct and falsifiable signature of the model in the strong-field regime.

C.4 Gravitational Wave Speed — The GW170817 Event

The most stringent observational bound comes from the binary neutron star merger GW170817, which imposes $|c_g/c - 1| \leq 10^{-15}$ [?]. In the linear regime of tensor perturbations, the anchorage tensor does not couple to purely transverse modes:

$$\left. \frac{\delta S_{\text{int}}}{\delta h_{\mu\nu}} \right|_{\mathcal{O}(h)} = 0,$$

so the wave equation reduces to the standard form $\square h_{\mu\nu} = 0$, yielding $c_g = c$ exactly. The model thus satisfies the strictest known observational constraint, clearing the hurdle that invalidates many modified-gravity theories.

C.5 Falsifiability Criterion and Kinematic Test

The formalism strictly preserves cosmological homogeneity and isotropy on large scales: the background expansion follows the Friedmann equations with $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The Hubble Tension emerges as a purely local effect, dependent on the observer’s peculiar velocity relative to the CMB rest frame. The direct refutation criterion is:

$$\lim_{|v_{\text{pec}}| \rightarrow 0} H_0^{\text{local}}(v_{\text{pec}}) = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

If, after analytically correcting for each observer’s peculiar velocity, the observed dispersion in H_0 remains unchanged, the model is falsified. This test can be performed using local Type Ia supernova catalogs segmented by motion relative to the Cosmic Microwave Background.

ACKNOWLEDGMENTS

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Appendix C: References

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